SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I T.karimulla/192465046 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

17/20

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

Given

- IP Address : 192.168.10.65
- Subnet Mask : 255.255.255.192 (/26)
- Default Gateway : 192.168.10.1

1. Subnet identification

- Subnet Mask : /26
- Block size : 64
- Subnets :
  - 192.168.10.0 - 63
  - 192.168.10.64 - 127
  - 192.168.10.128 - 191
  - 192.168.10.192 - 255

The IP 192.168.10.65 belongs to 192.168.10.64/26

2. Network Details

- Network Address : 192.168.10.64
- Broadcast Address : 192.168.10.127
- Valid Host Range : 192.168.10.65 - 192.168.10.126

3. IP and Gateway Verification

- IP Address (192.168.10.65) : Valid
- Gateway (192.168.10.1) : Invalid (belongs to subnet 192.168.10.0

4. Configuration Error

- Default Gateway is in a different subnet

5. Optimized Configuration

- IP Address : 192.168.10.65
- Default Gateway : 192.168.10.126

6. Total Usable Hosts

- Formula $= 2^{(32-26)} - 2$
- $= 64 - 2 = 62$ usable hosts

② Given

- Device A: 2001:db8::FF00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

1. Expanded IPv6 Addresses

Devices A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix

Device A

2001:db8:0:0::/64

Device B

2001:db8:0:0::/64

3. Same Subnet?

Yes. Both belong to 2001:db8:0:0::/64

4. Addressing Error

The IPv6 addresses are correctly configured. Since they are the same subnet, the communication problem is likely caused by another issue such as:

- IPv6 disabled
- Firewall blocking traffic
- Interface down
- Incorrect routing

Device A:

2001:db8:0:0::10/64

Device B:

2001:db8:0:0::20/64.

6. Available Hosts

A/64 Subnet Providers

2^{64} host addresses ≈ 18,446,744,073,709,551,616

(3) Given Network

192.168.1.0/24

Allocated using:

- /26
- /27
- /28

Subnet details

Subnet 1 (/26)

- Network: 192.168.1.0
- Broadcast: 192.168.1.63
- Hosts: 192.168.1.1 - 192.168.1.62
- Usable Hosts: 62

Subnet 2 (/27)

- Network: 192.168.1.64
- Broadcast: 192.168.1.95
- Hosts: 192.168.1.65 - 192.168.1.94
- Usable Hosts: 30

Subnet 3 (/28)

- Network: 192.168.1.96
- Broadcast: 192.168.1.111
- Hosts: 192.168.1.97 - 192.168.1.110
- Usable hosts: 14

Total unused Addresses : 144

Inefficiant Allocation

- Some departmetnts receive more addresses than required
- Other department experience address shortages
- Address space is wasted

Optimized VLSM Recommendation:

Department	Hosts Needed	Subnet
Dept A	60	/26
Dept B	30	/27
Dept C	14	/28
Small Departments	6	/29

(4)

Expanded Addresses

- Device A:
  2001:0db8:0000:0000:0000:ff00:0042:8329
- Device B:
  2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix

- 2001:db8:0:0::/64

Same Subnet ?

- Yes

Configuration issue

- IPV6 addressing is valid; investigate interface status firewall or routing

Optimized Addresses:

. Device A: 2001:db8:0:0::10/64

. Device B: 2001:db8:0:0::20/64

Available Hosts

. 2^{64} addresses

⑤ Routing table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination IP:

192.168.2.100

1. Logest Prefix Match

Matching route:

192.168.2.0/24

2. Next-Hop Router

10.0.0.3

3. Destination Verification

The IP 192.168.2.100 belongs to 192.168.2.0/24

4. Routing Error

If packets are not reaching the distination, possible issues include

- Incorrect next-hop Configuration
- Interface failure
- Missing route on the next Router
- Incorrect Subnet Configuration

5. Optimized Routing Table

Destination	Next hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

6. If No Matching Route Exists
The router forwards the packet using the Default Route (10.0.0.1).